

Unitary representations

$$G = \mathbb{R}^N$$

$$T = \mathbb{Z}^N$$

$$\sqrt{G} = \mathbb{T}^N = \mathbb{R}^N / \mathbb{Z}^N$$

$$T \backslash G$$

$$G \simeq \mathbb{R}^N = L^2(\mathbb{R}^N / \mathbb{Z}^N, d\theta)$$

$$L^2(T \backslash G, d\mu)$$

$$(t, \varphi) \mapsto (\pi(t)\varphi : x \mapsto \varphi(x+t))$$

$$\overline{SL_N(\mathbb{R})}$$

lie gr. locally qut grp

$$SL_N(\mathbb{R}) = \{g \in M(N, \mathbb{R}) \mid \det g = 1\}$$

$$g \text{ left action } \lambda_g(x) = gx$$

$$\text{right action } f_g(x) = xg^{-1}$$

(Haar measure) essentially unique

lem: 1) $\exists$ borel measure ν on G , finite on compact, s.t.

$$(\lambda_g)_* \nu = \nu$$

$$2). \forall g \in G, (\ell_g)_* \nu = \nu$$

pf. let λ be left-invariant measure on $M(N, \mathbb{R})$.

λ is λ_g, ℓ_g -inv. for $g \in G$, change variable

$SL_N(\mathbb{R}) = \{N \times N \text{ matrix with } \det = 1\}$

Then

$$d_G(gx, gy) = d_G(x, y)$$

left G-invariant metric

$\Rightarrow$

$$d_G(x, y) \leq d_G(x, z) + d_G(z, y)$$

$$= \log(1 + \|id - xz^{-1}\|) + \log(1 + \|id - zy^{-1}\|)$$

$$\text{where } id - xz^{-1} - zy^{-1} + xy^{-1} = (id - xz^{-1})(id - zy^{-1})$$

$$\leq \log(1 + \|id - xz^{-1}\| + \|id - zy^{-1}\| + \|id - xz^{-1} - zy^{-1} + xy^{-1}\|)$$

$$\leq \log(1 + \|id - xz^{-1}\| + \|id - zy^{-1}\| + \|id - xz^{-1} - zy^{-1} + xy^{-1}\|)$$

$$\log(1 + \|id - xy^{-1}\|)$$

Triangle inequality

$$d_G(x, y) = \log(1 + \|id - xy^{-1}\|) \leq \log(1 + \|id - xz^{-1}\| + \|id - zy^{-1}\| + \|id - xz^{-1} - zy^{-1} + xy^{-1}\|)$$

$$x \in M(N, \mathbb{R}), \text{ operator norm } \|x\| = \sup_{v \neq 0} \frac{\|xv\|}{\|v\|}, \text{ } \| \cdot \| \text{ such that}$$

Metric: (ad hoc) if we use Lie theory, more simple.

□

right invariant is similar

$$= \lambda(y, \det \leq 2, \frac{\det(y)}{y} \in A) = \nu(A)$$

$$= \lambda(x, \det \leq 2, \frac{\det(g^{-1}x)}{g^{-1}x} \in A)$$

$$\nu(gA) = \lambda(x \in M(N, \mathbb{R}), \frac{\det(x)}{x} \in gA)$$

$$1 \leq \det x \leq 2$$

$$1. \text{ Define } \nu(A) = \lambda\{x \in M(N, \mathbb{R}) : \frac{\det(x)}{x} \in A\}$$

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let T be a discrete subgr of G . Then $\exists \epsilon > 0$.

$$s.t. \quad B_G(\epsilon, 2\epsilon) \cap T = \{e\}$$

The quotient space $X = T \backslash G$ is the set of left cosets $x^{-1}Tg$.

$$\text{Quotient metric: } d_{T \backslash G}(x, x') = \inf_{g \in T} d_G(xg, x'g)$$

let $p: G \rightarrow X$ be the projection map. $\epsilon = \epsilon(1g)$

lem: $A \subseteq G$, the map p induces a homeo. from

$$B(1g, \epsilon) \rightarrow B(p(1g), \epsilon). \text{ And } p^{-1}B(p(1g), \epsilon)$$

is a disjoint union of balls $B(1g, \epsilon)$.

$$\text{Th. let } \epsilon(1g) = \frac{1}{2} \inf_{g' \neq 1g} d(1g, g')$$

$$= \frac{1}{2} \inf_{g' \neq 1g} d(1g, g^{-1}g')$$

$g^{-1}Tg$ discrete subgr

$$d(1g', 1g) = d(1g, (g'^{-1}g)) \geq \epsilon(1g)$$

$\Rightarrow B(1g', \epsilon), B(1g, \epsilon)$ disjoint.

lem. $\exists$ a local measure μ on X s.t. right G -inv

$$i.e. \quad A \subseteq G, \quad \mu(A) = \mu(A(1g))$$

pf: First notice ~~$SL_n(\mathbb{Q})$ is dense in $SL_n(\mathbb{R})$~~ $SL_n(\mathbb{R})$ is second countable. $A \subseteq SL_n(\mathbb{Q})$

$$\exists \epsilon(1g) \text{ s.t. } B(1g, \epsilon(1g)) \rightarrow B(p(1g), \epsilon(g)) \text{ homeo.}$$

~~we get a countable covering of~~ $X = T \backslash G \subset \bigcup B(p(1g), \epsilon(1g))$

$\exists B_i$ in the countable basis of G

Then get a family $G_i \subset B_i$, each B_i injective $g \in B_i \subset B(1g, \epsilon(1g))$

Def: A discrete subgroup Γ of G is a lattice if

(4) $\mu(\Gamma(G)) < \infty$. compact (uniform) if $\Gamma(G)$ compact.

Examples. (1) $SL_N(\mathbb{Z})$ is a non-cocompact lattice of $SL_N(\mathbb{R})$

(2) $SL_N(\mathbb{Z})$ also contains cocompact lattices

Unitary representations.

Def: let G be a locally cpt gr and $\mathcal{H}$ a Hilbert space

let $U(g) \in \mathcal{H}$ be the group of unitary operators on $\mathcal{H}$.

That is $U \in L(\mathcal{H})$. linear operators on $\mathcal{H}$ s.t

$$U^* U = U U^* = \text{id.} \quad (\text{i.e. } (U(v), U(w)) = (v, w))$$

$$Uv \in \mathcal{H}$$

Def: A unitary rep of G on $\mathcal{H}$ is a morphism

$$\pi: G \rightarrow U(\mathcal{H}) \quad \text{s.t. the map } v \mapsto Uv \in \mathcal{H}$$

$G \mapsto \pi(g)v$ is continuous from $G \rightarrow \mathcal{H}$

Example. $\mathcal{H} = L^2(\Gamma \backslash G, d\mu)$ $G \curvearrowright \mathcal{H}$ $Uv \in \mathcal{H}$

$$(\pi(g)U)(x) = U(gx)$$

is a unitary representation

Def. $\pi(g)$ preserves the measure μ . s.t

$$\langle \pi(g)U, \pi(g)U \rangle = \int \overline{U(xg)} U(xg) d\mu(xg) = \int \overline{U(x)} U(x) d\mu(x) = \langle U, U \rangle$$

continuity. $\forall \varphi \in C_c(X)$. s.t. $\varphi_n \xrightarrow{L^2} \varphi$ (X not loc cpt Hausdorff, μ -bnd)

$$\| \pi(g) \varphi - \pi(g) \varphi_n \|_{L^2} \leq \| \varphi - \varphi_n \|_{L^2} + \| \pi(g) \varphi_n - \pi(g) \varphi \|_{L^2}$$

$$\| \pi(g) \varphi_n - \pi(g) \varphi \|_{L^2} \xrightarrow{m \rightarrow \infty} 0$$

dominated convergent
uniform continuity

$$d(\varphi, g, x) = \inf (d(g, g_0), d(g, g_1)) \leq d(g, g_0, g_1) = d(g, g)$$

So $\sup \pi(g) \varphi_n$ is contained in $\sup \pi(g) \varphi_n$

a neighborhood of $\sup(\pi(g) \varphi_n)$ with size $d(g, g)$.

All functions $(\pi(g_m) \varphi_n)_m$ support on a same cpt set. $\square$

Example: $L^2(X, \mu) = \{ \varphi \in L^2(X, \mu) \mid \int \varphi d\mu = 0 \}$

G invariant subspace of $\mathcal{H}$.

and $L^2(X, \mu)^G = \{ 0 \}$, G -invariant function

By definition. $\forall g \in G$. $\varphi(gx) = \varphi(x)$ for a.e. x

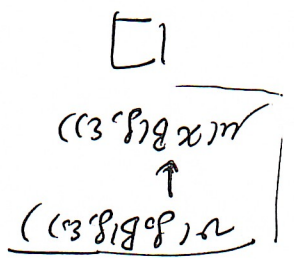
Then for μ -a.e. x $\varphi(gx) = \varphi(x)$ for ν -a.e. g

Fix a such x . then $\varphi(gx) \neq \varphi(x)$ for a g in a set

of ν measure 0.

Then such xg also has μ measure 0

Hence $\varphi(y) = \varphi(x)$ for μ -a.e. y .



$\square$

⑥

Def. Matrix coefficient for $v, v' \in \mathcal{H}$.

Let $g \in G$ and $g \in G$.

$$\langle \pi(g)v, v' \rangle = \langle \pi(g)v, v' \rangle$$

$$v, v' \in G.$$

$$\langle \pi(g)v, v' \rangle = \langle \pi(g)v, v' \rangle$$

If $\pi(h)v = v$, $\pi(h)v' = v'$, then

$$\langle \pi(g)v, v' \rangle = \langle \pi(g)v, v' \rangle = \langle \pi(g)v, \pi(h)v' \rangle$$

$$= \langle \pi(g)v, v' \rangle$$

Ergodicity of quotients of $SL_2(\mathbb{R})$

For $t > 0$, and $s \in \mathbb{R}$, let $a_t = \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}$, $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$

$$u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$$

Thm. Let T be a lattice in $SL_2(\mathbb{R})$. $t \neq 0$ ($s \neq 0$)

Then the action of a_t (u_s or u_s) is ergodic

for the measure μ .

Consequence of the proposition

Thm. For any vector v in $\mathcal{H} = L^2(X, \mu)$.

v is at $(u_s$ or $u_s)$ -inv. then v is

G -invariant

Lemma. (Mautner) $\forall v \in \mathcal{H}$ let $G_v = \{g \mid \pi(g)v = v\}$

i). G_v is a closed subgroup

$$ii). \quad g \in G_v \Leftrightarrow \langle \pi(g)v, v \rangle = \|v\|^2$$

(7)

with $\lim g_n = g$ $\lim s_n s'_n = e$

Then $g \in G_v$

by continuity

1)

Pf.

$$|| \langle C_v v, g \rangle | = | \langle \pi(g) v, v \rangle | \leq ||v||^2$$

$$||\pi(g)v - v||^2 = 2||v||^2 - 2\operatorname{Re}(\pi(g)v, v)$$

$$\pi(g)v = v \Rightarrow \operatorname{Re}(\pi(g)v, v) = ||v||^2 \Rightarrow \langle \pi(g)v, v \rangle = ||v||^2$$

$\Leftrightarrow$ trivial

$$!!!) \quad \pi(g)v = \lim \pi(g_n)v$$

by

$$\begin{aligned} \lim \pi(g_n s'_n) v &= \pi(e)v = v \\ \langle C_v v, g \rangle &= \lim \langle C_v v, g_n \rangle = \lim \langle C_v v, s_n g_n s'_n \rangle \\ &= \langle C_v v, e \rangle = ||v||^2 \end{aligned}$$

pf of Prop:

$$\text{let } g_n = u_s \quad s'_n s_n = a_{t-n} \quad s'_n = a_{t-n}$$

$$\begin{pmatrix} t \\ t^2 \end{pmatrix} \begin{pmatrix} 1 \\ s \end{pmatrix} \begin{pmatrix} 1 \\ t^{-1} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ s t^{-2} \end{pmatrix}$$

$$\text{Suppose } t > 0. \quad s_n g_n s'_n = a_{t-n} u_s a_{t-n} = u_s t^{-2n} \xrightarrow{n \rightarrow \infty} e$$

By Lem $u_s \in G_v$. u_s sim. larly

As $t \in R$

考试内容到 mixing 为止, 后面为拓展内容, 不同类型的动力系统

Hence $G_n = G$

There three subgroups generate $SL(2, \mathbb{R})$.

At, u_s, u_{-inv} for $\forall t \in \mathbb{R} > 0, s \in \mathbb{R} \in G_n$

Finally. For $at - inv$ (u_s or u_{-inv}), we get

Then
$$S_n g_{S_n'} = \begin{pmatrix} 1 & \frac{-t+1}{snp} \\ 0 & 1 \end{pmatrix} \xrightarrow{n \rightarrow \infty} \textcircled{0}$$

$$S_n' = u_{s-np} = \begin{pmatrix} 1 & -snp \\ 0 & 1 \end{pmatrix} \quad S_n = u_{snp} = \begin{pmatrix} 1 & snp \\ 0 & 1 \end{pmatrix}$$

let
$$g_n = \begin{pmatrix} p/q & \frac{-t+1}{snp} \\ 0 & p/q \end{pmatrix} \quad g = at$$

pf. It is sufficient to prove $t = \frac{p}{q}$ rational

$u_{s-inv} \Rightarrow at - inv \text{ for } \forall t > 0$

⑧